



Introduction

To

Device Configuration



CONFIGURING CISCO DEVICES: IOS BASICS

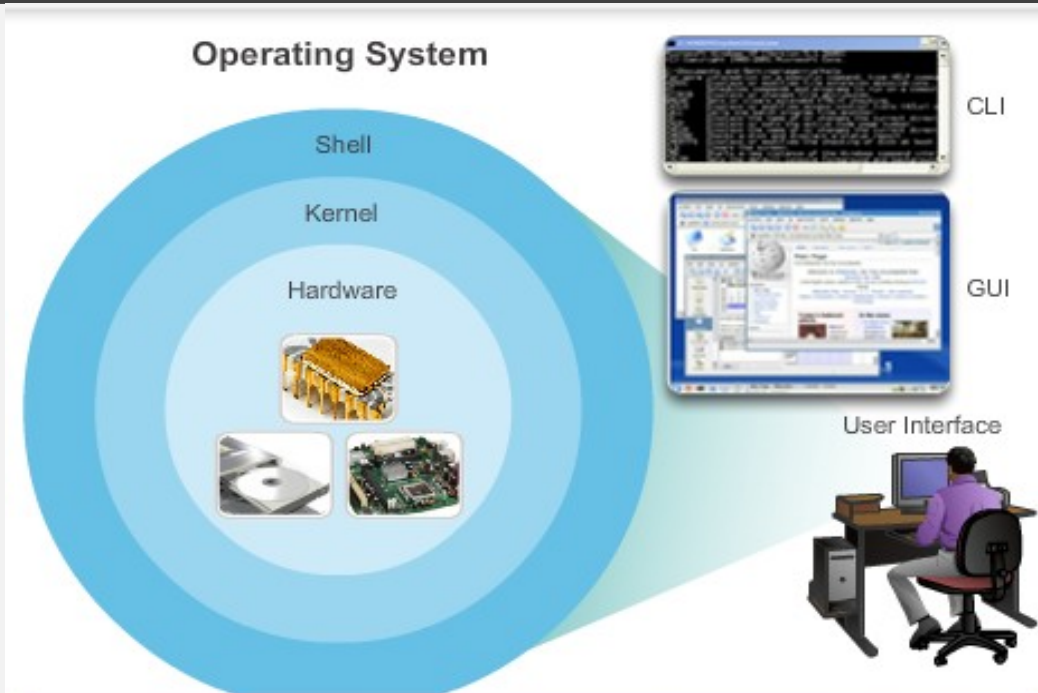
❑ **What is Network for you?**

❑ **What are network device ?**

- Similar to a personal computer, a router or switch cannot function **without an operating system.**
- Without an operating system, the hardware has no capabilities. **Cisco IOS** is the system software in **Cisco devices.**
- The Cisco IOS(Internetwork Operating System) is used for most Cisco devices:-
 - Routers
 - Switches
 - Wireless access points

- IOS provides devices with the following **network services**:
 - Basic routing and switching functions
 - Reliable and secure access to networked resources
 - Network scalability
- The IOS **stored in none volatile** memory area called **flash**
- The IOS can be upgraded to **newer versions** or to have new features added.
- The IOS is **copied into RAM** when the device is **powered on**, and the **IOS runs from RAM** when the device is operating.

CISCO IOS OPERATING SYSTEMS



Shell: The user interface that allows users to request specific tasks from the computer. These requests can be made either through the CLI or GUI interfaces.

Kernel: Communicates between the hardware and software of a computer and manages how hardware resources are used to meet software requirements.

Hardware: The physical part of a computer including underlying electronics.

PURPOSE OF OS

- PC operating systems perform technical functions that enable
 - Use of a mouse
 - View output
 - Enter text
- Switch or router IOS provides options to
 - Configure interfaces
 - Enable routing and switching functions
- All networking devices come with a **default IOS**
- **Possible to upgrade** the IOS version or feature set

IOS FLASH

- Non-volatile storage – not lost when power is lost
- Can be changed or overwritten as needed
- Can be used to store multiple versions of IOS
- IOS copied from flash to volatile RAM



IOS FUNCTIONS

- Major functions performed or enabled by Cisco routers and switches include:



CONFIGURATION FILES

- Network devices depend on two types of software for their operation:
 - Operating system and
 - Configuration.
- Like the operating system in any computer, the operating system facilitates the basic operation of the device hardware components.
- Configuration files contain the Cisco IOS Software commands used to customize the functionality of a Cisco device.
- Commands are parsed (translated and executed) by the Cisco IOS Software when the system is booted (from the startup-config file) or when commands are entered in the CLI while in configuration mode.

- A network administrator creates a configuration that defines the desired functionality of a Cisco device.
- A Cisco network device contains two configuration files:
 - **The running configuration file:** Used during the **current operation** of the device.
 - **The startup configuration file:** Used as the backup configuration and loaded when the device is started.
- A configuration file can be stored remotely on a server as a backup.

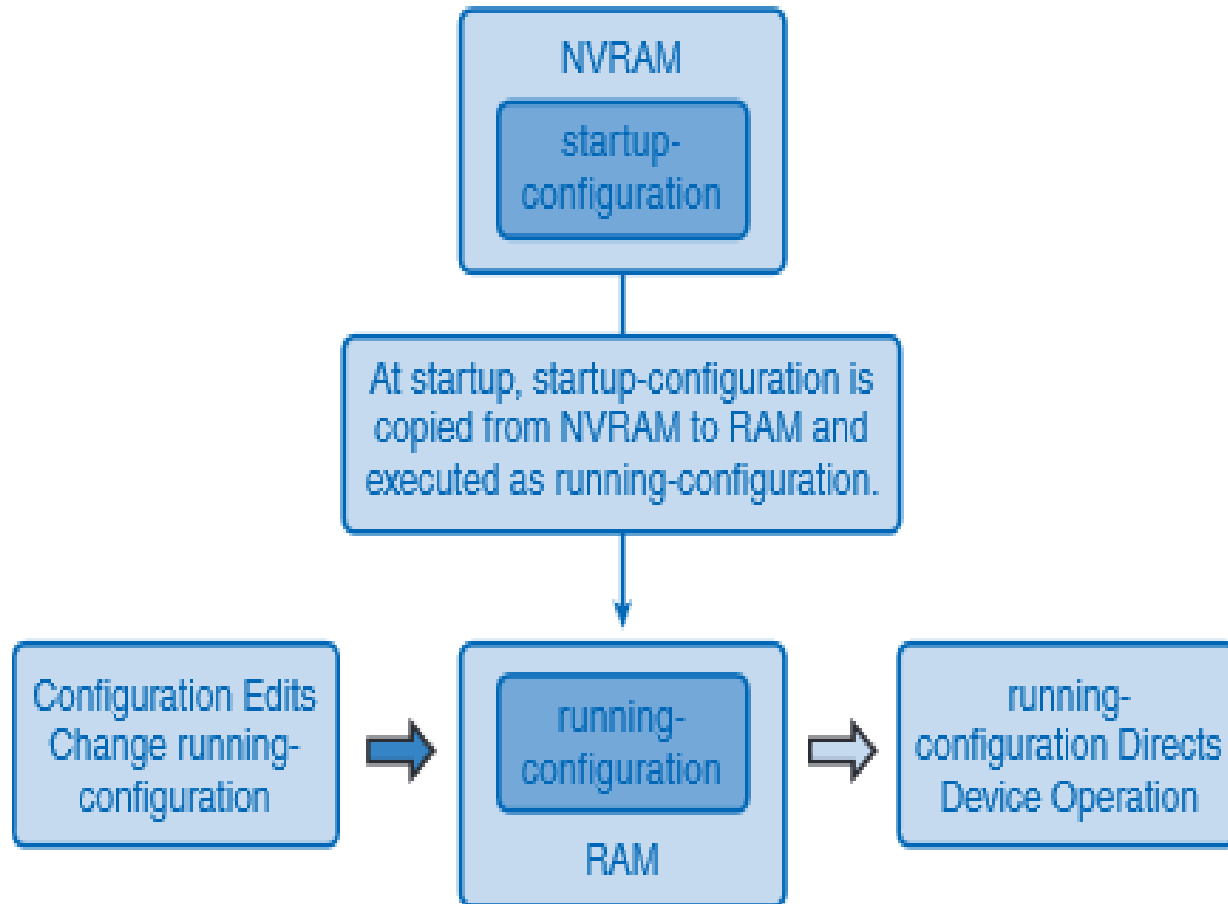
- The startup configuration file (startup-config) is used during system startup to configure the device.
- The startup configuration file or startup-config file is stored in **non-volatile RAM (NVRAM)**
- Because NVRAM is nonvolatile, when the Cisco device is turned off, the file remains intact.
- The startup-config files are loaded into RAM each time the router is started or reloaded.
- After the configuration file is **loaded** into RAM, it is considered the **running configuration or running-config**.

RUNNING CONFIGURATION

- The running configuration is **modified** when the network administrator performs device configuration.
- Changes to the running configuration will immediately **affect the operation** of the Cisco device.
- After making any changes, the administrator has the option of **saving** those changes to the **startup-config file** so that they will be used the next time the device restarts.
- Because the running configuration file is in RAM, it is **lost if the power** to the device is turned off or if the device is restarted.
- **Changes made** to the running-config file will also be lost if they are **not saved to the startup-config** file before the device is powered down

FIGURE 11-2 SHOWS THE RELATIONSHIP BETWEEN THE CONFIGURATION FILES

Figure 11-2 Configuration Files



NAVIGATING THE IOS

CISCO IOS MODES OF OPERATION

IOS Mode Hierarchical Structure

User EXEC Command-Router>

ping
show (limited)
enable
etc.

Privileged EXEC Commands-Router#

all User EXEC commands
debug commands
reload
configure
etc.

Global Configuration Commands-Router(config)#

hostname
enable secret
ip route

interface ethernet
serial
dsl
etc.

Interface Commands-Router(config-if)#

ip address
ipv6 address
encapsulation
shutdown/ no shutdown
etc.

router rip
ospf
eigrp
etc.

Routing Engine Commands-Router(config-router)#

network
version
auto summary
etc.

line vty
console
etc.

Line Commands-Router(config-line)#

password
login
modem commands
etc.

NAVIGATING THE IOS PRIMARY MODES

User EXEC Mode

Limited examination of router.
Remote access.

```
Switch>  
Router>
```

The **User EXEC** mode allows only a limited number of basic monitoring commands and is often referred to as view-only mode.

Privileged EXEC Mode

The **Privileged EXEC** mode, by default, allows all monitoring commands, as well as execution of configuration and management commands.

Detailed examination of router. Debugging and testing. File manipulation. Remote access.

```
switch#  
Router#
```

NAVIGATING THE IOS GLOBAL CONFIGURATION MODE AND SUBMODES

Privileged EXEC Mode

Privileged EXEC Mode

Detailed examination of router, Debugging and testing.
File manipulation. Remote access.

Switch#
Router#



Global Configuration Mode

Global configuration commands.

Switch(config)#
Router(config)#



Other Configuration Modes

Specific service or interface configurations.

Switch(config-mode)#
Router(config-mode)#

IOS Prompt Structure

```
Router>ping 192.168.10.5
```

```
Router#show running-config
```

```
Router(config)#Interface FastEthernet 0/0
```

```
Router(config-if)#ip address 192.168.10.1 255.255.255.0
```

The prompt changes to denote the current CLI mode.

```
Switch>ping 192.168.10.9
```

```
Switch#show running-config
```

```
Switch(config)#Interface FastEthernet 0/1
```

```
Switch(config-if)#Description connection to WEST LAN4
```

NAVIGATING THE IOS

NAVIGATING BETWEEN IOS MODES

Router con0 is now available.

Press RETURN to get started.

User Access Verification

Password:

Router>

User-Mode Prompt

Router>**enable**

Password:

Router#

Privileged-Mode

Router#**disable**

Router>

User-Mode Prompt

Router>**exit**

Router

NAVIGATING THE IOS

NAVIGATING BETWEEN IOS MODES (CONT.)

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.
End with CNTL/Z.
Switch(config)#interface vlan 1
Switch(config-if)#exit
Switch(config)#exit
Switch#
```

```
Switch#configure terminal
Enter configuration commands, one per line.
End with CNTL/Z.
Switch(config)#vlan 1
Switch(config-vlan)#end
Switch#
```

```
Switch#configure terminal
Enter configuration commands, one per line.
End with CNTL/Z.
Switch(config)#line vty 0 4
Switch(config-line)#interface fastethernet 0/1
Switch(config-if)#end
Switch#
```

THE COMMAND STRUCTURE

CONTEXT SENSITIVE HELP

Context Sensitive Help

```
Switch#cl?  
clear clock
```

Command options - display a list of commands or keywords that start with the characters **cl**

```
Switch#clock set ?  
hh:mm:ss Current Time
```

Command explanation - the IOS displays what command arguments or variables can be next, and provides an explanation of each

```
Switch#clock set 19:50:00 ?  
<1-31> Day of the month  
MONTH Month of the year
```

Command explanation with more than one argument or variable option

```
Switch#clock set 19:50:00 25 June 2012  
Switch#
```

THE COMMAND STRUCTURE COMMAND SYNTAX CHECK

```
Switch#>clock set  
% Incomplete command.  
Switch#clock set 19:50:00  
% Incomplete command.
```

The IOS returns a help message indicating that required keywords or arguments were left off the end of the command.

```
Switch#c  
% Ambiguous command: 'c'
```

The IOS returns a help message to indicate that there were not enough characters entered for the command interpreter to recognize the command.

```
Switch#clock set 19:50:00 25 6  
^  
% Invalid input detected at '^'  
marker.
```

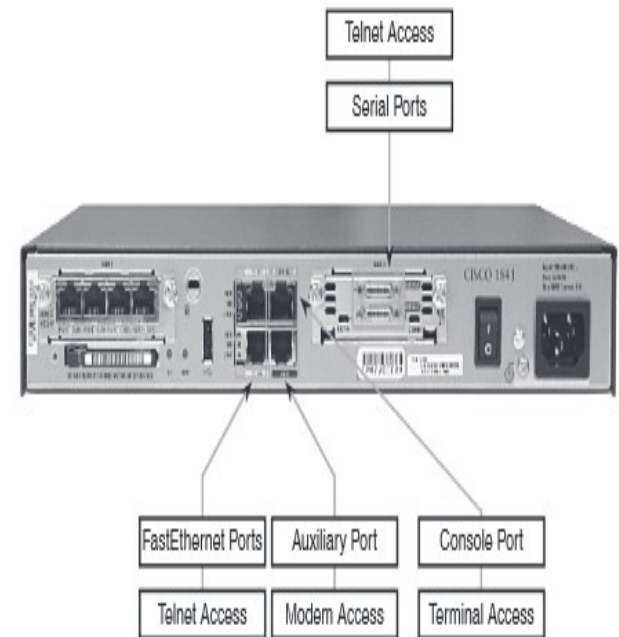
The IOS returns a "^" to indicate where the command interpreter can not decipher the command.

ACCESS METHODS

- You can access the CLI environment in several ways.
- The most common methods are
 - **Console**
 - **Telnet or SSH**
 - **AUX port**



Figure 11-1 Accessing the Cisco IOS on a Device



CONSOLE ACCESS METHOD

□ Console port

- Device is accessible even if **no networking services** have been configured (out-of-band)
- Need a special **console cable**
- Allows **configuration commands** to be entered
- Should be configured **with passwords to prevent unauthorized access**
- Device should be located in a secure room so **console port can not be easily accessed**



- The CLI can be accessed through a console session.
- A console used **directly connect a computer or terminal** to the **console port** on the **router or switch**.
- The **console port is a management port** that provides out-of-band access to a router.
- The console port is often used to access a device when the networking services **have not been started** or **have failed**.

❑ Console use include the following:

- The initial configuration of the network device
 - Disaster recovery procedures and troubleshooting where remote access is not possible
 - Password recovery procedures
-
- When a router is **first placed** into service, networking parameters have **not yet been configured**. Therefore, the router cannot communicate through a network.
 - To prepare for the **initial startup and configuration**, a computer running **terminal emulation software** is connected to the **console port** of the device.

❑ Telnet

- A method **for remotely accessing** a CLI session is to telnet to the router.
- Unlike the console connection, Telnet sessions require **active networking** services on the device.
- The network device must have at least **one active interface**.
- A host with a **Telnet client** can access the **virtual teletype interface** (vty) sessions running on the Cisco device.
- For security reasons, the IOS requires that the Telnet session use a **password**, as a minimum authentication method.

❑ Secure Shell (SSH)

- The Secure Shell (SSH) protocol is a **more secure** method for remote device access.
- This protocol provides the structure for a **remote login** similar to Telnet, except that it utilizes more **secure network services**.
- SSH provides **stronger password authentication** than Telnet and uses **encryption** when trans-ported session data.
- As a best practice, always use SSH in place of Telnet whenever possible.

❑ **AUX**

- Another way to establish a CLI session remotely is through a **telephone dialup connection** using a modem connected to the router auxiliary (AUX) port.
- Similar to the console connection, this method **does not require** networking services to be configured or available on the device.
- The **console port** is also preferred over the **AUX port** for troubleshooting because it **displays** router startup, debugging, and error messages by default.

TELNET, SSH, AND AUX ACCESS METHODS

Telnet

- Method for remotely accessing the CLI over a network
- Require active networking services and one active interface that is configured

Secure Shell (SSH)

- Remote login similar to Telnet but utilizes more security
- Stronger password authentication
- Uses encryption when transporting data

Aux Port

- Out-of-band connection
- Uses telephone line
- Can be used like console port

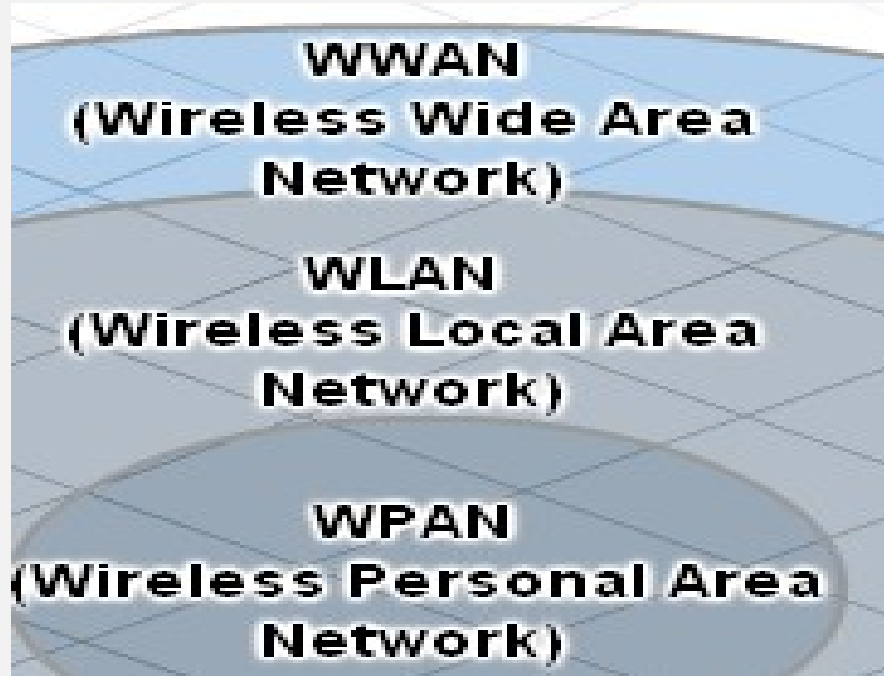


WIRELESS TECHNOLOGIES AND DEVICES

- In addition to the **wired network**, various technologies exist that allow the transmission of information between hosts **without cables**. These are known as **wireless technologies**.
- Wireless technologies use **electromagnetic waves** to carry information between devices.
- An electromagnetic wave is the same medium that carries **radio signals through the air**.

TYPES OF WIRELESS NETWORKS AND THEIR BOUNDARIES

- Wireless networks are grouped into three major categories:



WPAN

- This is the smallest wireless network used to connect various peripheral devices such as **mice, keyboards** to computer.
- All of these devices are dedicated to a **single host** with usually use **IR or Bluetooth technology**.

WLAN

- WLAN is typically used to extend the boundaries of the **local wired network (LAN)**.
- WLANs use **RF technology** and conform to the **IEEE 802.11** standards.
- They allow many users to connect to a wired network through a device known as an **Access Point (AP)**.

WWAN

- WWAN networks provide coverage over extremely large areas.
- A good example of a WWAN is the **cell phone network**.
- These networks use technologies such as **Code Division Multiple Access (CDMA)** or **Global System for Mobile Communication (GSM)** and are often **regulated by government agencies**.

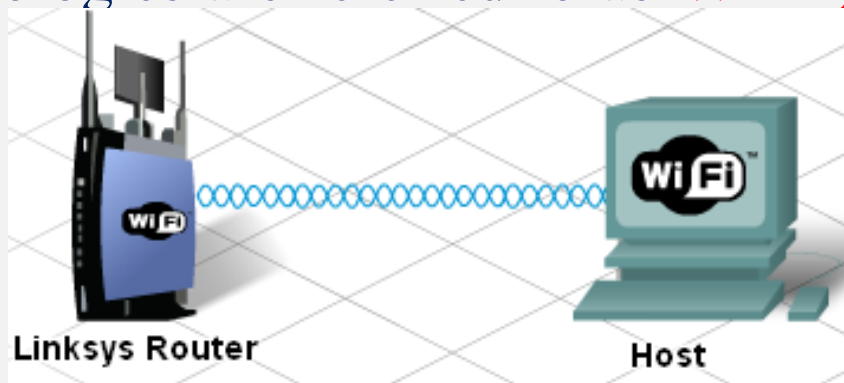
	WPAN	WLAN	WWAN
Standards	Bluetooth v2.0+ EDR**	IEEE802.11 a/b/g/n, HiperLAN, HiperLAN2	GSM, GPRS, CDMA
Speed	< 3 Mbps	1-540 Mbps	10-384 Kbps
Range	Short	Medium	Long
Applications	Peer-to-Peer device to device	Home, small business and enterprise networks	PDAs, mobile phones, cellular access

** EDR is Enhanced Data Rate

Speed and ranges are constantly increasing with newer technologies.

WIRELESS LAN STANDARDS

- A **number of standards** have been developed to ensure that wireless devices can communicate.
- They specify the **RF spectrum** used, **data rates**, how the **information is transmitted**.
- The main organization responsible for the creation of wireless technical standards is the **IEEE**.
- The currently available amendments are 802.11a, 802.11b, 802.11g and 802.11n. (802.11n is not ratified at the time of this writing.)
- Collectively these technologies are referred to as **Wi-Fi, Wireless Fidelity**.



Common IEEE WLAN Standards

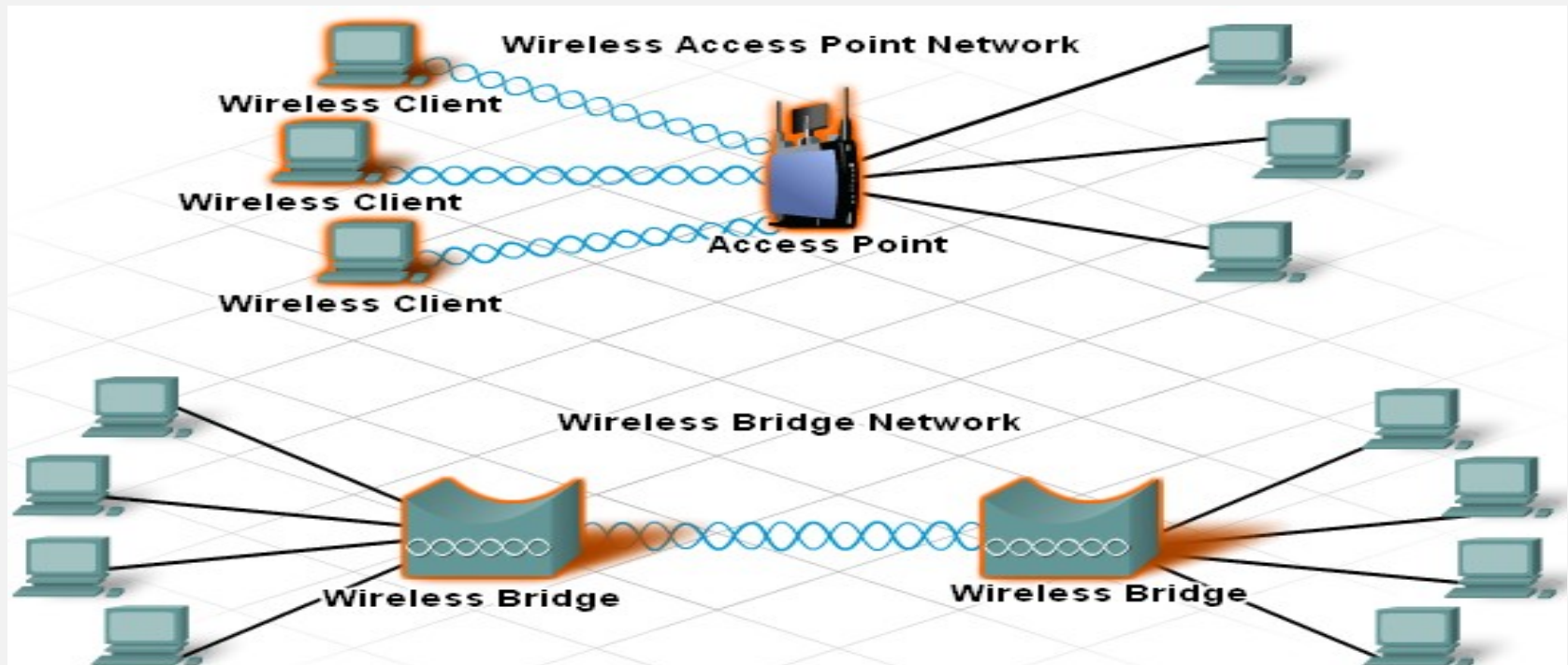
Standard	Release Date	Frequency	Data Rate (Max)	Maximum Range*
802.11	July 1997	2.4 GHz	2 Mbps	undefined
802.11a	October 1999	5 GHz	54 Mbps	50 m
802.11b	October 1999	2.4 GHz	11 Mbps	100 m
802.11g	June 2003	2.4 GHz	54 Mbps	100 m
**802.11n	Draft - Nov 2006 Release - Jan 2007 Approval - April 2007	2.4 GHz or 5 GHz	540 Mbps	250 m

*Maximum Range - This value can vary widely. ~ The 802.11n standard is still in draft and values may change.

WIRELESS LAN COMPONENTS

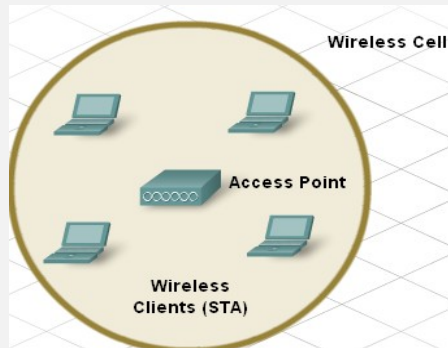
□ Antennas:

- Used on **APs** and **Wireless bridges**
- Increases the **output signal strength** from a wireless device.
- **Receives wireless signals** from other devices such as STAs.



- Antennas are classified according to the way they radiate the signal.
- **Directional antennas** concentrate the signal strength into one direction.
- **Omni-directional antennas** are designed to emit equally in all directions.
- By concentrating all of the signal into one direction, **directional antennas can achieve great transmission distances.**

- When building a wireless network, it is important that the wireless components connect to the **appropriate WLAN**. This is done using a Service Set Identifier (**SSID**).
- The SSID is a **case-sensitive**, alpha-numeric string that is up to **32-characters**.
- It is sent in the **header of all frames** transmitted over the WLAN.
- The SSID is used to **tell wireless devices which WLAN they belong to** and with which other devices they can communicate.



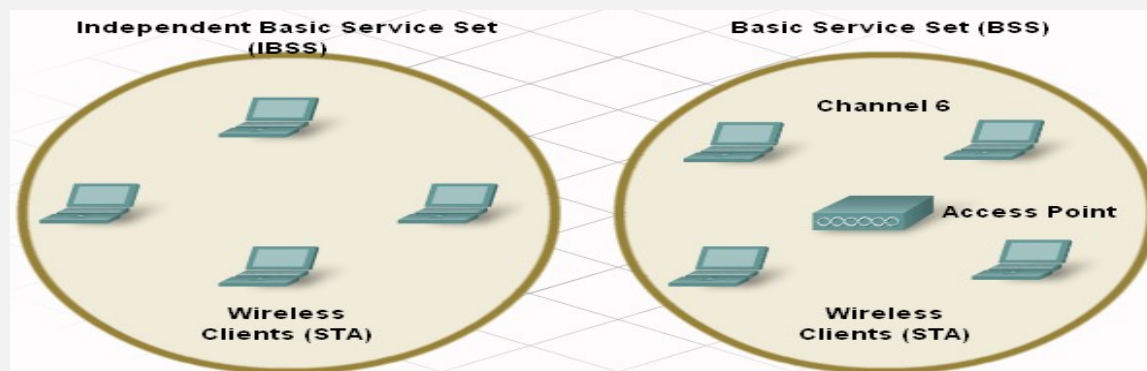
- There are two basic forms of WLAN installations: **Ad-hoc** and **infrastructure mode**.

Ad-hoc

- The simplest form of a wireless network is created by connecting two or more **wireless clients** together in **a peer-to-peer network**.
- A wireless network established in this manner is known as an ad-hoc network and **does not include an AP**.
- **All clients** within an ad-hoc network are **equal**.
- The area covered by this network is known as an **Independent Basic Service Set (IBSS)**.

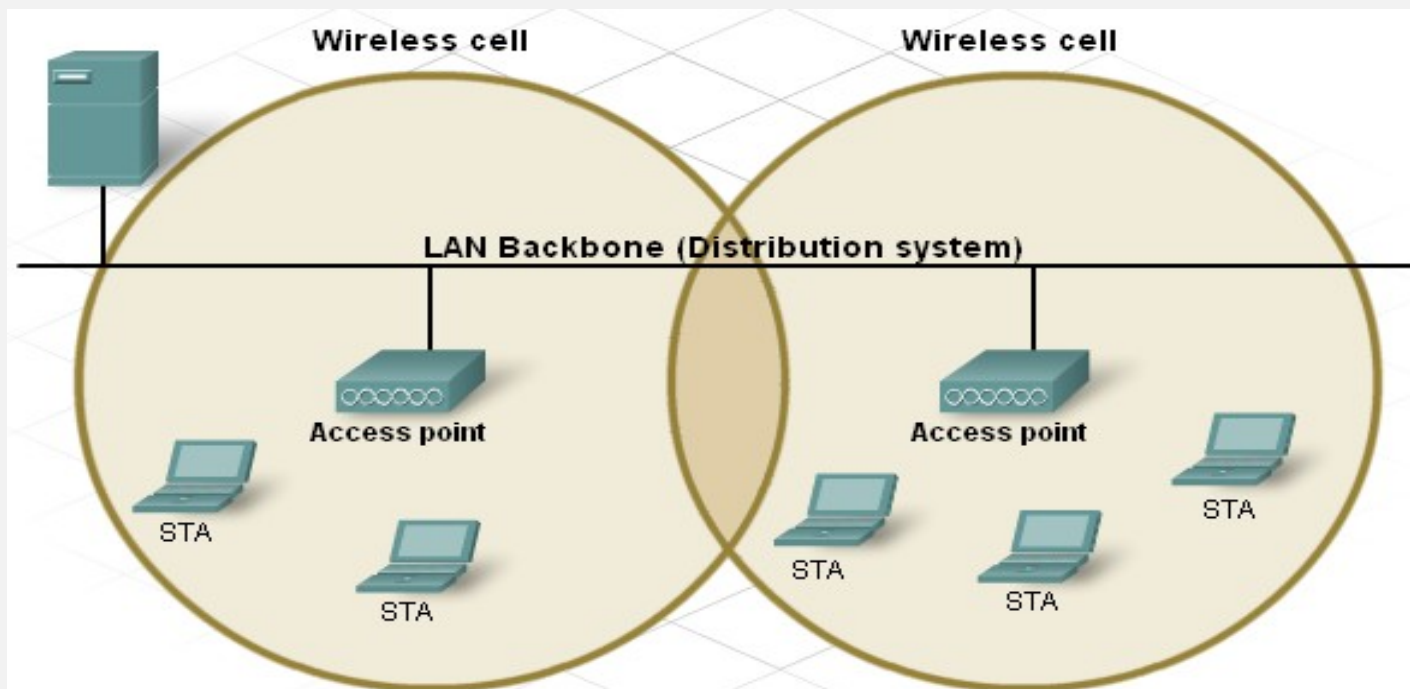
❑ Infrastructure Mode

- Infrastructure mode is the mode of wireless communication most often used in the **home** and **business environment**.
- In this form of WLAN, individual STAs **can not communicate directly with each other**. To communicate, each device must obtain **permission from the AP**.
- The **AP controls** all communications and ensures that all STAs have **equal access to the medium**.
- The area covered by a **single AP** is known as a **Basic Service Set (BSS) or cell**.

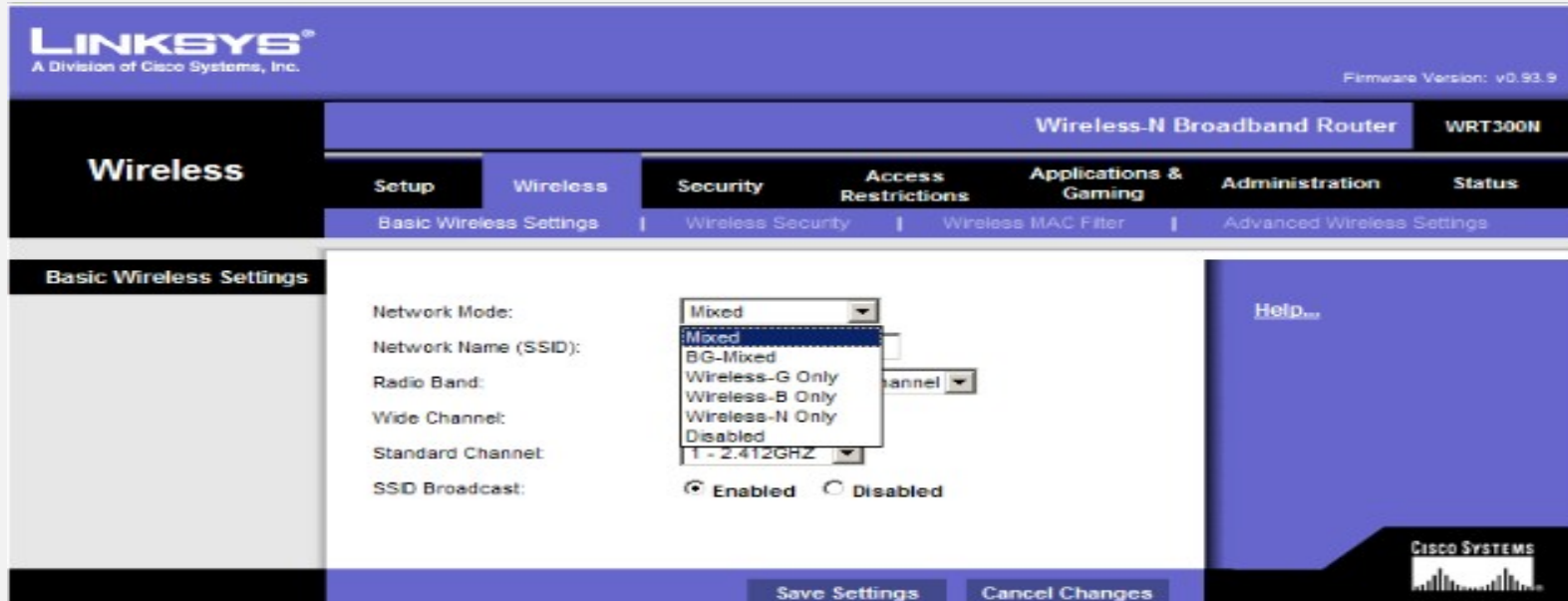


- The Basic Service Set (BSS) is the smallest building block of a WLAN.
- The area of coverage of a single AP is limited.
- To expand the coverage area, it is possible to **connect multiple BSSs** through a **Distribution System (DS)**.
- This forms an Extended Service Set (ESS).
- **An ESS uses multiple APs.** Each AP is in a separate BSS.

- In order to allow movement between the cells **without the loss of signal**, BSSs must overlap by **approximately 10%**.
- This allows the client to connect to the second AP before disconnecting from the first AP.



- Most integrated routers offer both **wired and wireless connectivity** and serve as the AP in the wireless network.
- Basic configuration settings such as **passwords, IP addresses, and DHCP settings** are the same whether the device is being used to connect **wired or wireless hosts**.
- When using the wireless functionality of an integrated router, **additional configuration** parameters are required, such:
 - **wireless mode,**
 - **SSID, and**
 - **wireless channels** to be used.



Wireless Mode

- Most home AP devices can support various modes, mainly **802.11B**, **802.11G** and **802.11N**.
- The type of mode enabled on the AP depends on the type of host connecting to it.
- If **only one type** of host connects to the AP device, set the mode to support it. If **multiple types of hosts** will connect, select **Mixed mode**.

SSID

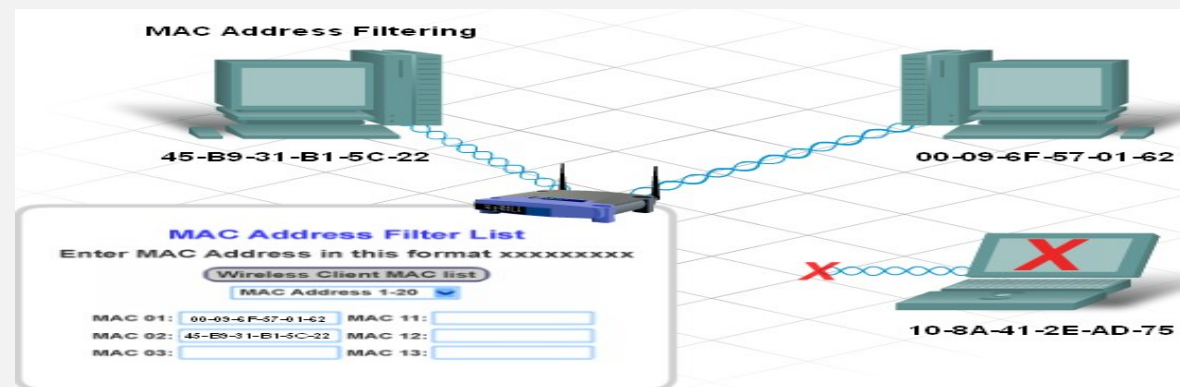
- The SSID is used to **identify** the WLAN. All devices that wish to participate in the WLAN must use the same SSID.
- To allow easy detection of the WLAN by clients, the SSID is **broadcast**.
- It is possible to **disable the broadcast feature** of the SSID. If the SSID is not broadcast; wireless clients will need to have this **value manually configured**.

Wireless Channel

- The choice of channel for an AP must be made relative to the other wireless networks around it.
- Adjacent BSSs must use **non-overlapping channels** in order to optimize throughput.

MAC Address Filtering

- MAC address filtering uses the MAC address to identify which devices are **allowed to connect to the wireless network**.
- When a wireless **client attempts to connect**, or associate, with an AP it will send MAC address information.
- If MAC filtering is enabled, the wireless router or **AP will look up its MAC address** a preconfigured list.
- Only devices whose MAC addresses have been prerecorded in the router's database will be allowed to connect.

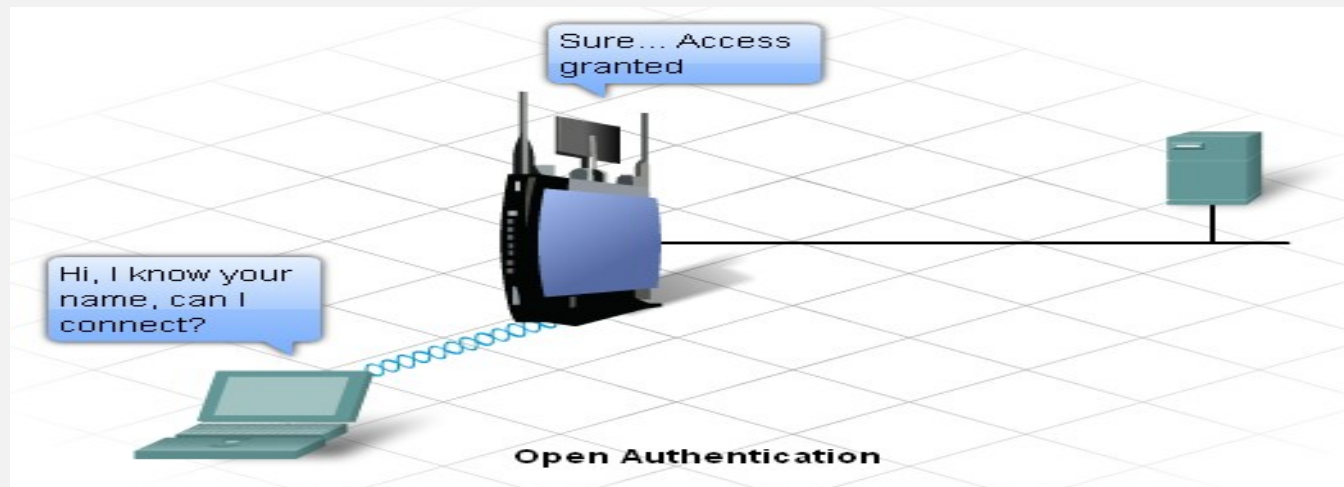


AUTHENTICATION ON A WLAN

- Another way to control **who can connect** is to implement authentication.
- Authentication is the process of permitting entry to a network based on a set of credentials.
- There are three types of wireless authentication methods:
 - open authentication,
 - PSK and
 - EAP.

❑ Open Authentication

- By default, wireless devices do not require authentication.
- Any and all clients are able to associate regardless of who they are. This is referred to as open authentication.
- Open authentication should only be **used on public wireless** networks such as those found in many schools and restaurants.



- **Pre-shared keys (PSK)**
- With PSK both the AP and client must be configured with the same **key or secret word**.
- ✓ The **AP sends a random string of bytes** to the client.
- ✓ The **client accepts the string, encrypts it** (or scrambles it) based on the key, and **sends it back to the AP**.
- ✓ The **AP gets the encrypted** string and uses its key to decrypt (or unscramble) it.
- ✓ If the **decrypted string received** from the **client matches** the original string sent to the client, the client is **allowed to connect**.
- PSK performs **one-way authentication**, that is, the host authenticates to the AP.

❑ Extensible Authentication Protocol (EAP)

- EAP provides **mutual, or two-way, authentication** as well as user authentication.
- The **backend server** functions separately from the **AP** and maintains a **database** of valid users that can access the network.
- When using **EAP**, the user, **must provide a username and password** which is checked against the RADIUS database for validity. If valid, the user is authenticated.



ENCRYPTION ON WLAN

- Authentication and MAC filtering **may stop an attacker from connecting to a wireless network** but it will **not prevent them from being able to intercept** transmitted data.
- Since **there are no distinct boundaries** on a wireless network, and all traffic is transmitted through the air, it **is easy for an attacker to intercept, or sniff the wireless frames**.
- **Encryption** is the process of transforming data so that even if it is intercepted it is **unusable**.
- Protocol that used in Wireless technology to Encrypt are:-
 - **WEP**
 - **WPA**

Wired Equivalency Protocol (WEP)

- Wired Equivalency Protocol (WEP) is an **advanced security feature** that **encrypts network** traffic as it travels through the air.
- WEP uses **pre-configured keys to encrypt and decrypt** data.



Wi-Fi Protected Access (WPA)

- WPA also uses **encryption keys** from 64 bits up to 256 bits.
- However, WPA, unlike WEP, **generates new, dynamic keys** each time a **client establishes a connection** with the AP.
- For this reason, WPA is considered **more secure than WEP** because it is significantly more difficult to crack.

ROUTER CONFIGURATION

COMMAND LINE INTERFACE

- Using the CLI to configure and monitor a device is very **different than using the SDM** (security device manager).
- The CLI does not provide **step-by-step** configuration assistance; therefore, it requires more planning and **expertise** to complete
- The **command history** is enabled by default and the system records ten command lines in the **history buffer**.
- To **recall** the most recent command in the history buffer, press **Ctrl-P** or the **Up Arrow key**. Repeat this process to recall successively older commands.
- To **return to a more recent** command in the history buffer, press **Ctrl-N** or the **Down Arrow key**. Repeat this process to recall successively more recent commands.

Command Shortcuts

- CLI recognizes partially-typed commands based on their first unique character. For example, type **int** instead of **interface**.
- **Tab** - Completes the remainder of a partially typed command or keyword
- **Ctrl-Z** - Exits configuration mode and returns to user EXEC
- **Ctrl-Shift-6** - Allows the user to interrupt **an IOS process** such as **ping** or **traceroute**.
- **Ctrl-C** - Aborts the current command and exits the configuration mode

COMMAND LINE INTERFACE

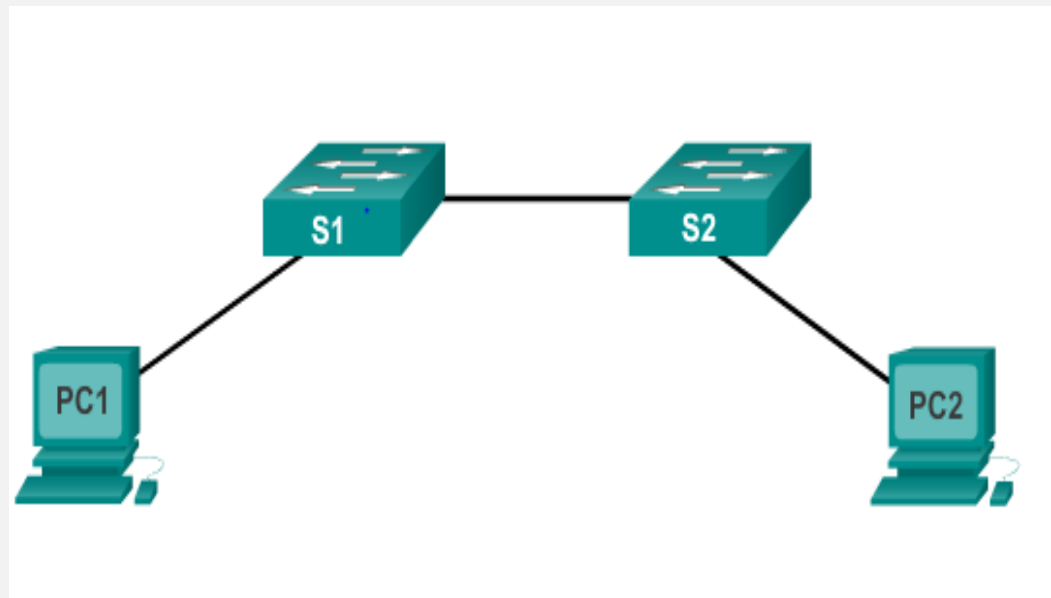
□ Using Show Commands

- The CLI enables a user to display relevant information about the configuration and operation of the device.
- To obtain this information, **show commands are used.**
- These commands are used to
 - View configuration files,
 - The status of the device interfaces and processes and
 - To verify the device operational status.
- Some of the more popular show commands are:
 - **Show running-config**
 - **Show interfaces**
 - **Show arp**
 - **Show ip route**
 - **Show users**
 - **Show version**

HOSTNAMES

Let's focus on

- Creating a two PC network connected via a switch
- Setting a name for the switch
- Configuring banner messages
- Saving the configuration

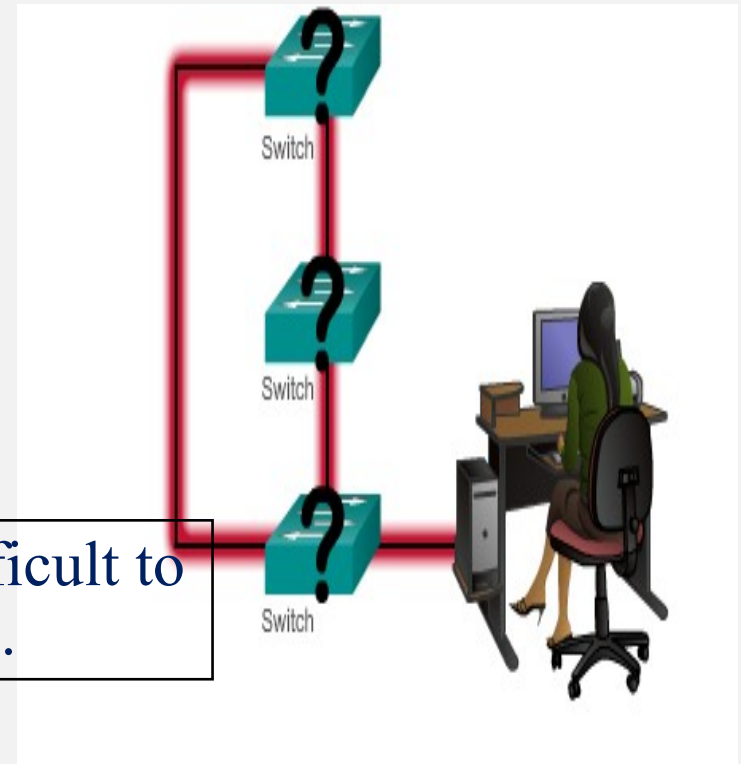


HOSTNAMES

Some guidelines for naming conventions are that names should:

- Start with a letter
- Should not Contain spaces
- End with a letter or digit
- Use only letters, digits, and dashes
- Be less than 64 characters in length

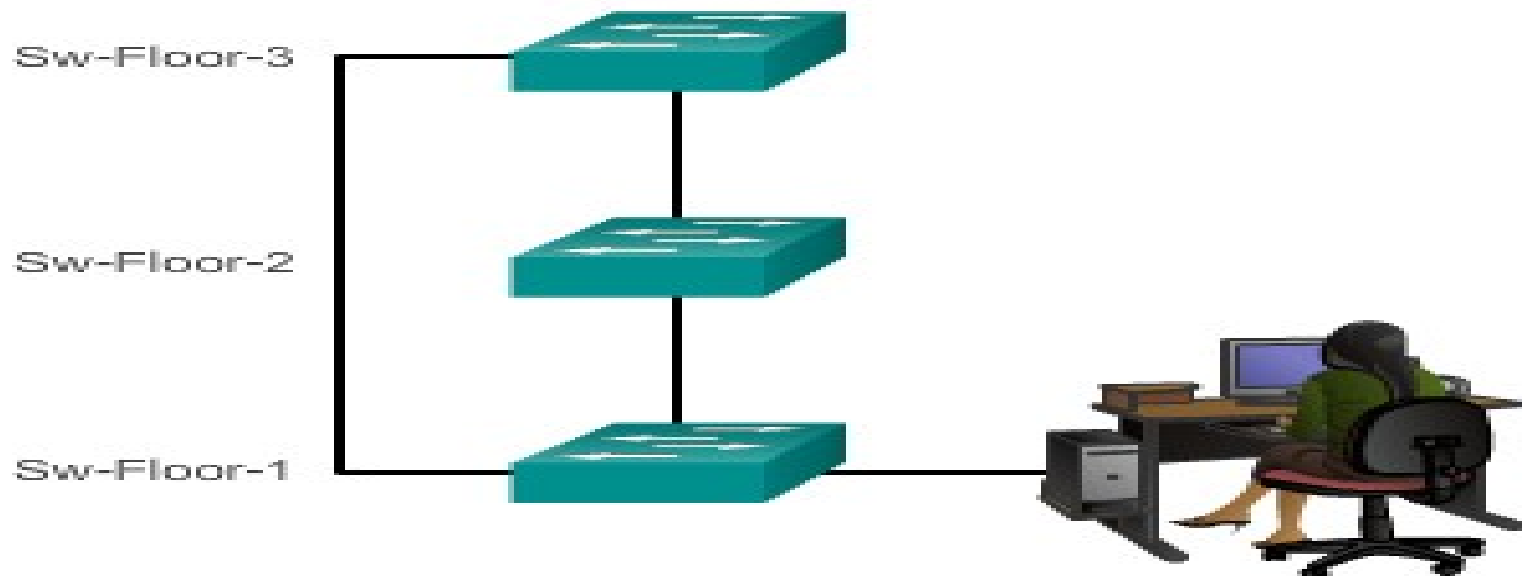
Without names, network devices are difficult to identify for configuration purposes.



HOST NAMES

Hostnames allow devices to be **identified** by network administrators over a network or the Internet.

Configuring Device Names



HOSTNAMES CONFIGURING HOSTNAMES

Configure a Hostname

Configure the switch hostname to be 'AtlantaHQSw1'.

```
Switch# configure terminal
```

Enter configuration commands, one per line. End with CNTL/Z.

```
Switch(config)# hostname AtlantaHQSw1
```

```
AtlantaHQSw1(config)#
```

You successfully configured the switch hostname.

SECURING DEVICE ACCESS

The passwords introduced here are:

- **Enable password** - Limits access to the **privileged EXEC mode**
- **Enable secret** - Encrypted, limits access to the privileged EXEC mode
- **Console password** - Limits device access using the console connection
- **VTY password** - Limits device access over **Telnet**

SECURING PRIVILEGED EXEC ACCESS

- use the **enable secret** command, not the older **enable password** command
- **enable secret** provides greater security because the password is **encrypted**.

```
Sw-Floor-1>enable
Sw-Floor-1#
Sw-Floor-1#conf terminal
Sw-Floor-1(config)#enable secret class
Sw-Floor-1(config)#exit
Sw-Floor-1#
Sw-Floor-1#disable
Sw-Floor-1>enable
Password:
Sw-Floor-1#
```

SECURING USER EXEC ACCESS

```
Sw-Floor-1(config)#line console 0
Sw-Floor-1(config-line)#password cisco
Sw-Floor-1(config-line)#login
Sw-Floor-1(config-line)#exit
Sw-Floor-1(config)#
Sw-Floor-1(config)#line vty 0 15
Sw-Floor-1(config-line)#password cisco
Sw-Floor-1(config-line)#login
Sw-Floor-1(config-line)#
```

- **Console port must be secured**
 - reduces the chance of unauthorized personnel physically plugging a cable into the device and gaining device access
- **vty lines allow access to a Cisco device via Telnet**

ENCRYPTING PASSWORD DISPLAY

service password-encryption

- prevents passwords from showing up as **plain text** when viewing the configuration
- purpose of this command is to keep **unauthorized individuals** from **viewing passwords** in the configuration file
- once applied, removing the encryption service does not reverse the encryption

Configuring Password Encryption

```
Enter the command to encrypt the plain text passwords.
Switch(config)# service password-encryption
Exit global configuration mode and view the running configuration.
Switch(config)# exit
Switch# show running-config
!
<output omitted>
!
line con 0
 password 7 094F471A1A0A
 login
!
line vty 0 4
 password 7 02095A0F024F28435B49150A1819
 login
!
!
end
```

BANNER MESSAGES

- Important part of the legal process in the event that someone is prosecuted for breaking into a device
- Wording that implies that a login is "**welcome**" or "**invited**" is not appropriate
- Often used for legal notification because it is displayed to all connected terminals

Limiting Device Access - MOTD Banner

```
LAB_A(config)#banner motd # This is a secure system. Authorized Access ONLY!!! #
```

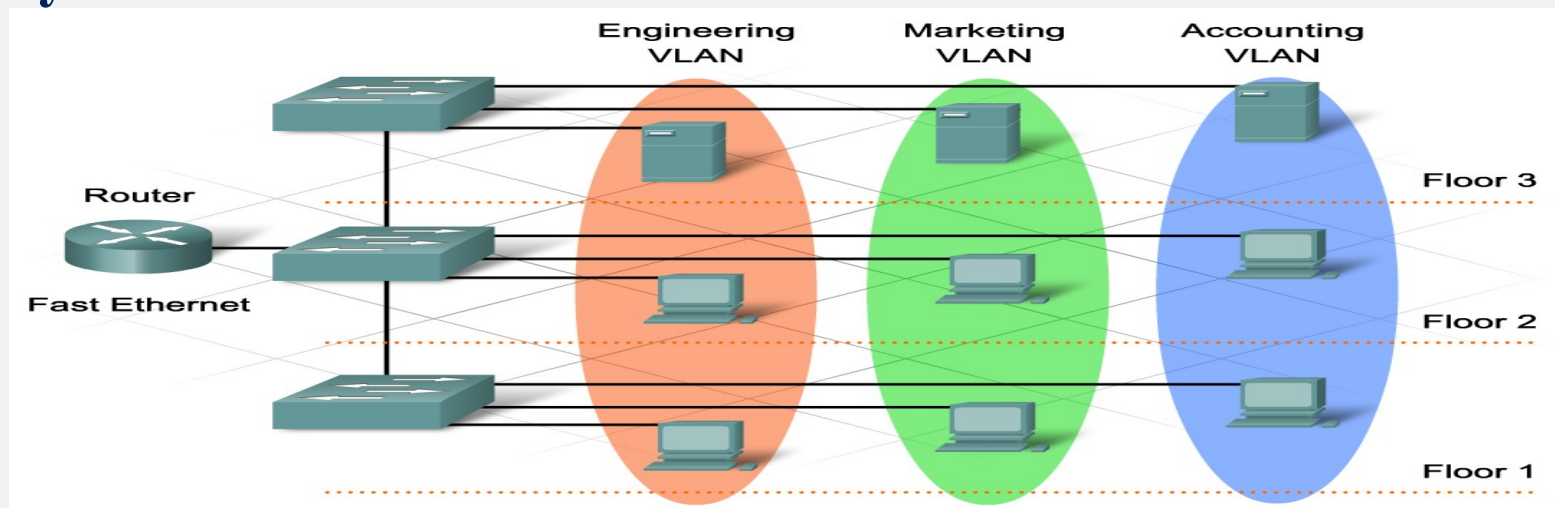
This configuration results in this message of the day banner.

Delimiting characters are not included in the message.

```
Sw1-Floor-1 con0 is now available
Press RETURN to get started.
This is a secure system. Authorized
Access ONLY!!!
User Access Verification
password:
Sw1-Floor-1>enable
Password:
Sw1-Floor-1#
```

VLAN

- In a switched network, virtual local area networks (VLANs) are created to **contain broadcasts** and **group hosts together** in communities of interest.
- A VLAN is a **logical broadcast domain** that can span multiple physical LAN segments.
- It allows an administrator to group together stations by **logical function**, by **project teams**, or **by applications**, without regard to physical location of the users.



VLAN

- A VLAN has two major functions:
 - ✓ A VLAN contains broadcasts.
 - ✓ A VLAN groups devices.
- In a switched network, a **device can be assigned** to a VLAN based on its location, **MAC** address, **IP** address.
- Administrators **assign** membership in a **VLAN** either **statically** or **dynamically**.
- **Static VLAN** membership requires an administrator to manually assign each switch port to a specific VLAN.
- As an example, **port fa0/3** may be assigned to **VLAN 20**. Any device that plugs into port fa0/3 automatically becomes a member of VLAN 20.

VLAN

- **Dynamic VLAN** membership requires a **VLAN management policy server (VMPS)**.
- The VMPS contains a **database** that maps **MAC addresses** to VLAN assignments.
- When a **device plugs** into a switch port, the **VMPS** searches the database and temporarily assigns that port to the appropriate VLAN.

- By default, VLAN1 is the **management VLAN**.
- An administrator will use the IP address of the **management VLAN** to configure the switch remotely.
- When a VLAN is created, it is assigned a **number and a name**.
-
- The VLAN number is any number from the range available on the switch, **except for VLAN1**.
- Some switches support approximately **1000 VLANs**; others support more than **4000**.
- Commands to create a VLAN using global configuration mode:
Switch(config)#vlan **vlan_number**
Switch(config-vlan)#name **vlan_name**
Switch(config-vlan)#exit

- Assign **ports** to be members of the VLAN.
- By default, all ports are **initially members of VLAN1**.
- Use the following commands to assign individual ports to VLANs:

Switch(config)#interface fa#/#

Switch(config-if)#switchport access vlan vlan_number

Switch(config-if)# exit

VLAN

- The following commands are used to verify and maintain VLANs:

show vlan

- Displays a detailed list of all of the VLAN numbers and names, along with the ports associated with each one.

show vlan brief

- Displays a summarized list showing only the active VLANs and the ports associated with each one

show vlan id id_number

- Displays information pertaining to a specific VLAN, based on ID number

VLAN

- **show vlan name** vlan_name
- Displays information pertaining to a **specific VLAN, based on name.**
- When a port is **disassociated** from a specific VLAN, it **returns to VLAN1.**

To **delete a VLAN**:

- Switch(config)#no vlan vlan_number

To disassociate a port from a specific VLAN:

- Switch(config)#interface fa##/##
- Switch(config-if)#no switchport access vlan **vlan_number**

SECURITY DEVICE MANAGER (SDM)

- In addition to **CLI**, other tools are available to assist in **configuring a router**.
- **Security Device Manager (SDM)** is a **graphical user interface (GUI)** device management tool.
- Unlike CLI, SDM can be used only for **in-band management tasks**.
- It uses a **step-by-step** approach to create a **basic router configuration** quickly and easily.
- Use the full SDM package to perform more advance configurations such as:
 - ✓ Configure additional LAN and WAN connections
 - ✓ Create firewalls
 - ✓ Configure VPN connections
 - ✓ Perform security tasks